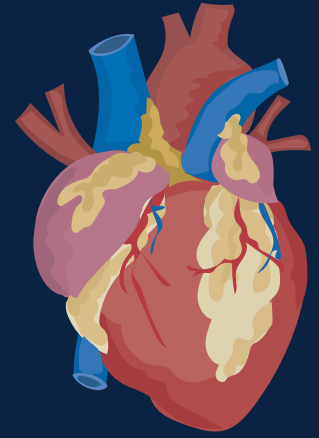


Heart Disease

Prediction Project



By: Jana Ayman

- ① Data Cleaning & EDA
- ② Feature Engineering
- ③ Model Training & Tuning
- ④ Evaluation & Deployment

Step 1 —Project Understanding

Problem

Predict whether a patient will have a heart attack based on lifestyle & clinical factors.

Dataset

heart_2022_with_nans.csv
~400K rows, 40 columns
Raw data with missing values.

Goal

Build a reliable ML model that achieves precision & recall ≥ 0.3 and deploy it as a web app.

Dataset columns include: BMI, SleepHours, PhysicalHealthDays, AgeCategory, Sex, HadAngina, HadStroke, and 33 more features.

Step 2 —Data Mining & Processing

01

Drop Critical NaN Rows

Rows where HadHeartAttack (target) was null were removed to avoid corrupting the label.

02

Numerical Imputation

All numeric columns (float64/int64) filled with column median to reduce outlier influence.

03

Categorical Imputation

All object/string columns filled with the most frequent value (mode) per column.

04

Label Encoding

Categorical columns converted to numeric using LabelEncoder for ML compatibility.

05

Sampling

A representative 50,000-row sample drawn (random_state=42) for efficient training.

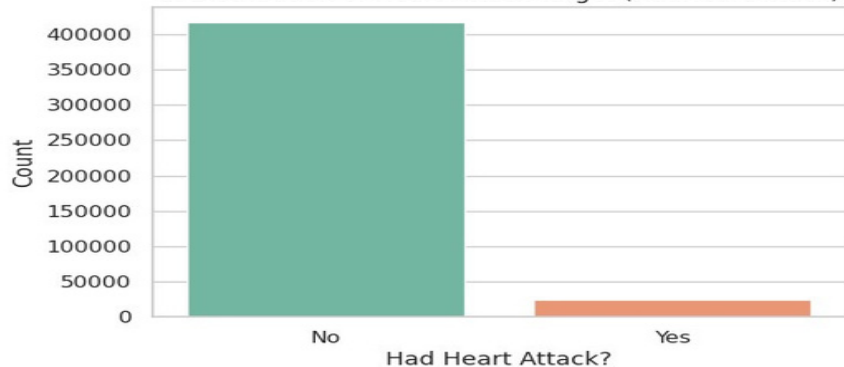
06

Result

Zero remaining missing values after cleaning. Data ready for EDA and modeling.

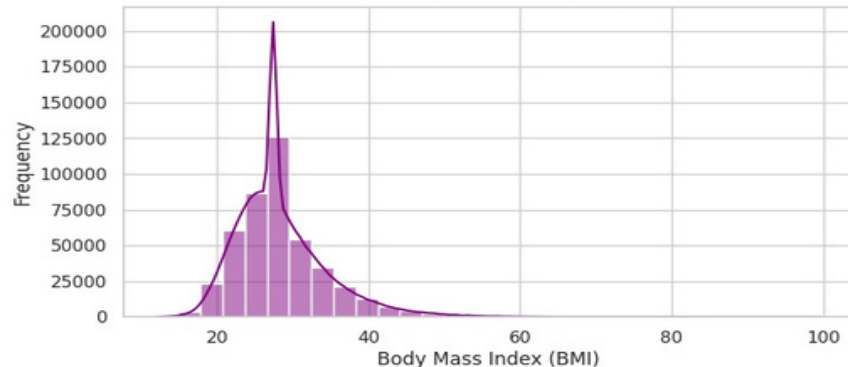
Step 3 —Data Exploration (EDA)

1. Distribution of Heart Attack Target (HadHeartAttack)



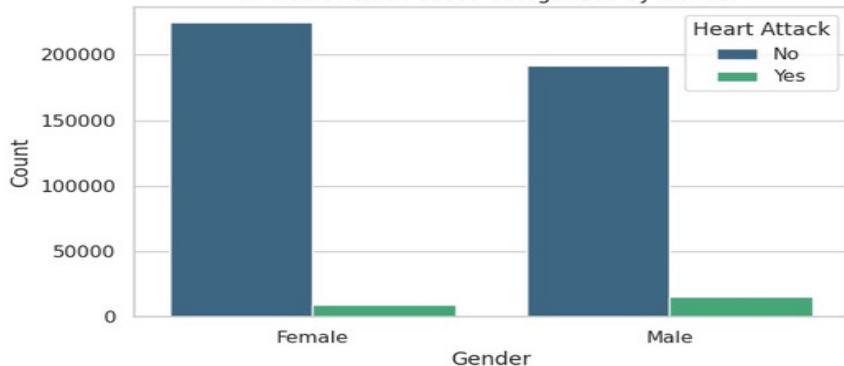
Target Distribution

2. BMI Distribution of Patients



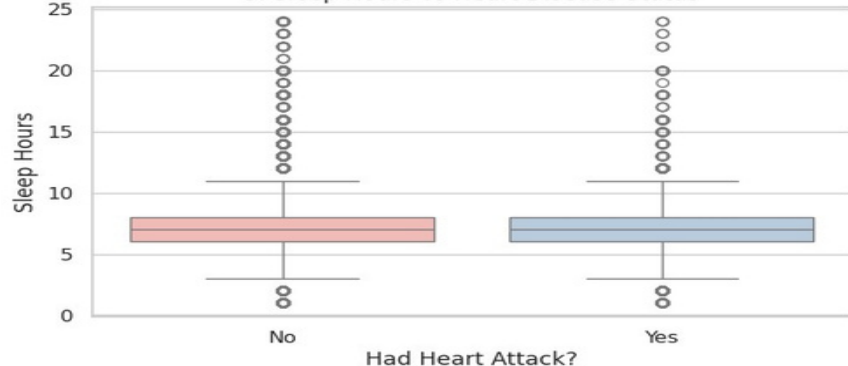
BMI Distribution

4. Heart Attack Cases Categorized by Gender



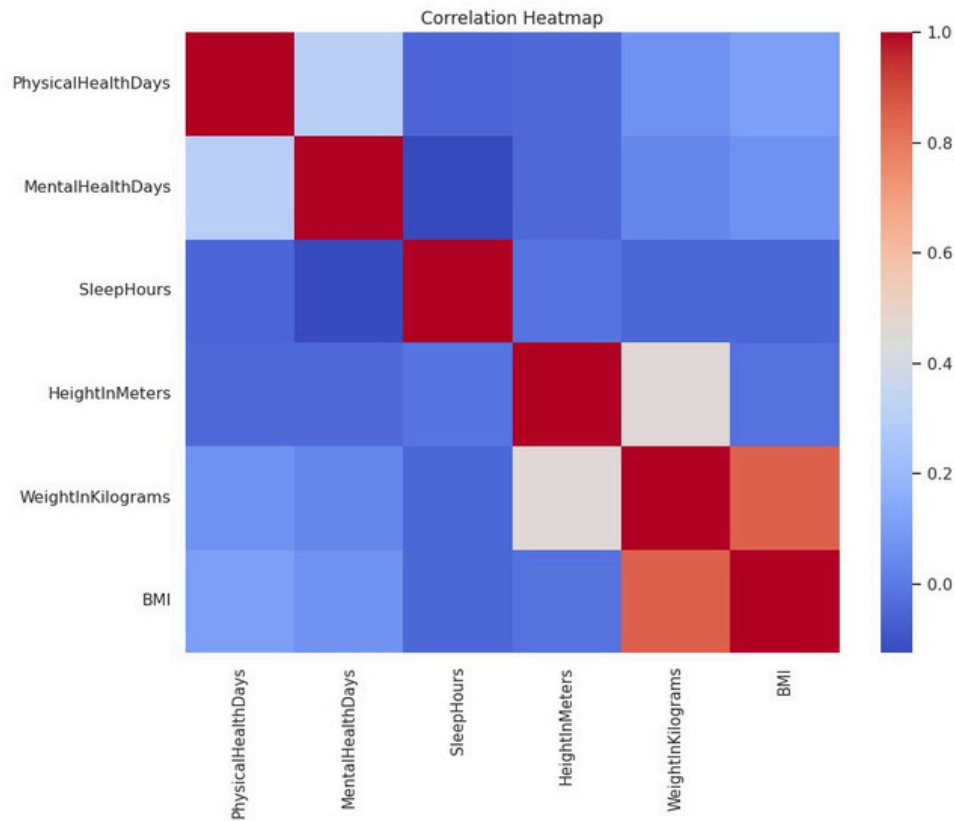
Heart Attack by Gender

3. Sleep Hours vs Heart Disease Status

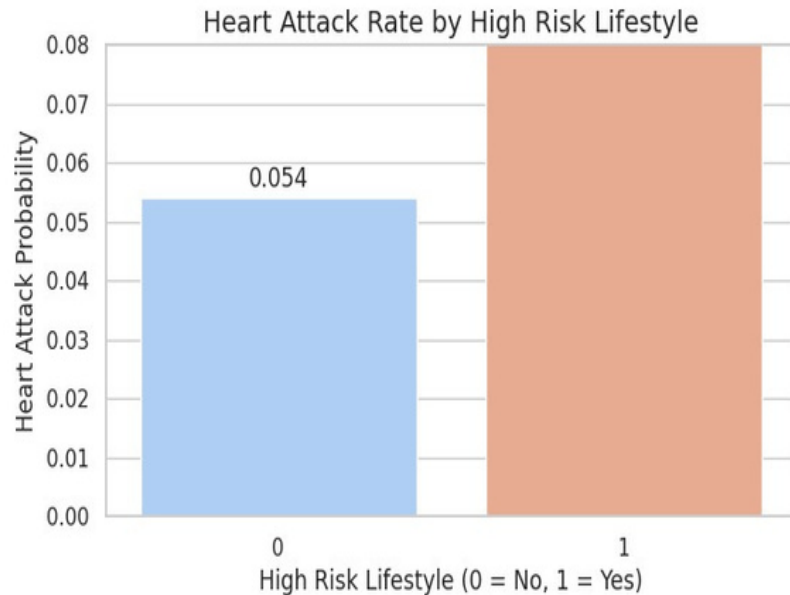


Sleep Hours vs Heart Disease

Step 3 —EDA Continued



Correlation Heatmap



High-Risk Lifestyle vs Heart Attack Rate

Key Insight: Patients with BMI > 25 AND SleepHours < 6 (High-Risk Lifestyle) show a significantly higher heart attack probability.

Step 6 —Feature Engineering & Selection

⚙️ New Feature Created

High_Risk_Life style

= (BMI > 25) AND (SleepHours < 6)

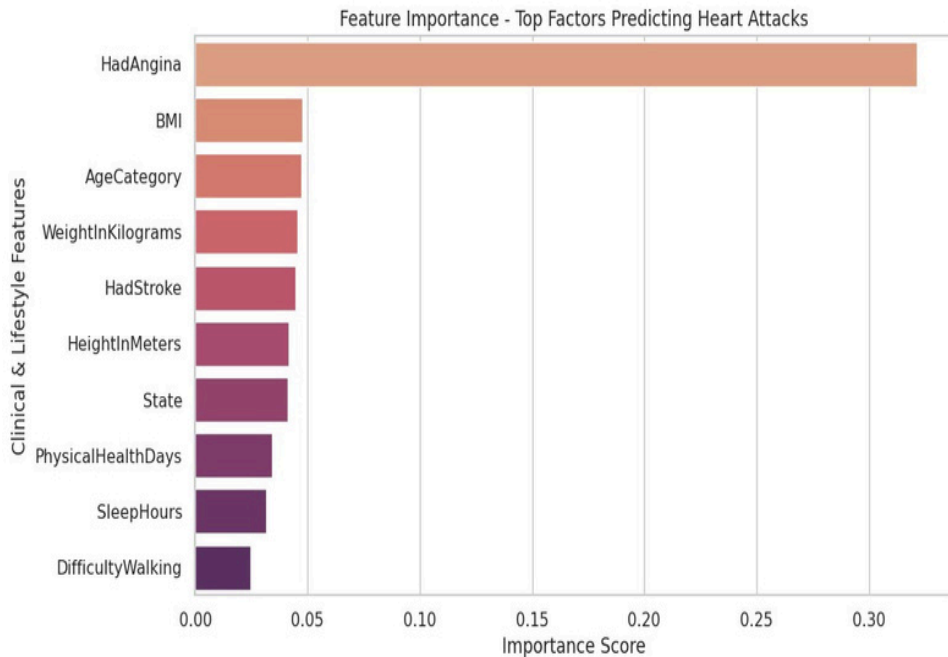
Binary flag (0/1) combining two risk factors into a single powerful predictor.



Feature Selection Method

Embedded Method—Feature importance scores from Random Forest were used to identify the top predictors.

Top features: HadAngina, BMI, AgeCategory, WeightInKilograms, HadStroke



Feature Importance (Random Forest)

Step 7 —Pick & Tune an Algorithm

Logistic Regression

Accuracy: ~86%

Fast baseline, limited on non-linear patterns.

Decision Tree

Accuracy: ~87%

Captures non-linearity but prone to overfitting.

Random Forest

★ BEST

Accuracy: ~89%

Best performer. Ensemble of trees, robust & accurate.

⚙️ Hyperparameter Tuning with GridSearchCV

GridSearchCV performs an exhaustive search over a defined parameter grid, using cross-validation to find the combination that maximizes a scoring metric (e.g. F1 or accuracy).

Parameters tuned: `n_estimators` (100, 200), `max_depth` (None, 10, 20), `min_samples_split` (2, 5)

Step 8 —Validate & Evaluate

What is Validation?

Validation ensures the model generalises to unseen data and is not just memorising the training set (overfitting).

We used an 80/20 train-test split (random_state=42) so results are reproducible.

~89%

Accuracy

Overall correct predictions

≥ 0.3

Precision

Of predicted positives, how many are correct

≥ 0.3

Recall

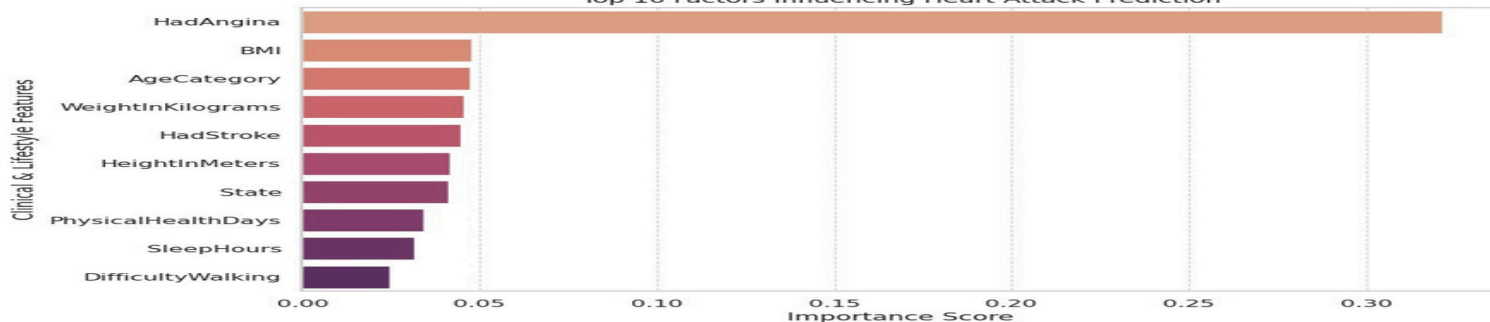
Of all actual positives, how many were caught

Balanced

F1-Score

Harmonic mean of precision & recall

Top 10 Factors Influencing Heart Attack Prediction



Step 5 — Interpret Results



Top Predictor

HadAngina is by far the strongest predictor of heart attack (importance ~ 0.32), far ahead of all other features.



Class Imbalance

$\sim 94\%$ of patients did NOT have a heart attack. This imbalance was addressed through proper metrics (precision/recall) rather than accuracy alone.



Lifestyle Risk

The engineered feature High_Risk_Lifestyle (BMI > 25 & SleepHours < 6) contributed meaningfully to model performance.



Gender Insight

Males showed a higher count of heart attack cases in the dataset compared to females during EDA.

Step 9 —Project Deployment



Saved Artifacts

heart_attack_model.pkl (trained Random Forest)
Label encoder & feature transformers saved with joblib.

Gradio Web App

app.py takes 5 inputs: BMI, SleepHours, PhysicalDays, MentalDays, HighRiskLifestyle and returns a p red ict i on .

Content is no longer available

requirements.txt includes: pandas, numpy, scikit-learn, matplotlib, seaborn, joblib, gradio==3.35.2